1 Composition

Summary

Menu	Material management → Composition → Composition
Transaction code	comp
Function authorization	comp

Usage

You use the "composition" application to control planning, monitoring and execution of the melting process. This application includes the following functions:

- Generation of charging orders
- Performance of composition planning based on existing and advised stocks as well as the planned bottom sump
- Reservation of the planned material for the planned melt
- Generation of a charging list
- Release for material provision (perform charging)
- Execution of composition based on the provided materials and the actual bottom sump
- Checking of the theoretical analysis
- Saving of theoretical analyses
- Display of the target analysis including tolerances and restrictions
- *Execution of charging and sampling at the terminal*
- Comparison of actual analysis and target analysis by identifying the elements using traffic light colors.
- Performance of re-composition

The configuration of composition functions are described in the relevant procedure.

Selection criteria

The following selection criteria are available in the application:

Category "charging order"

The selection criteria of this category have been designed to restrict data in the detail application "charging orders".

Order

Charging order

Order status

Order status of the charging order

Finished article

Finished article of the charging order

Machine

Planned melting aggregate

"Dates" category

The selection criteria of this category have been designed to restrict data in the detail application "charging orders".

Basic start date / basic end date

Temporal restriction using basic dates

Scheduled start/scheduled end

Temporal restriction by scheduling

Category "input material"

The selection criteria of this category have been designed to restrict data in the detail application "inventory of permitted materials".

Material

Input material.

Material designation

Designation of input material.

Material buffer

Material buffer to identify the material

Material type

Material type for classification

Category "options"

The selection criteria of this category have been designed to restrict data in the detail applications "inventory of permitted materials" as well as "charging orders".

Show additional material only

If this option is enabled, the detail application "inventory of permitted materials" will only show materials configured as "additional materials" for the output material (finished article of the charging order) that is to be produced by the charging order.

Show permitted material only

If this option is enabled, the detail application "inventory of permitted materials" will only show materials configured as "permitted input materials" for the material that is to be produced by the charging order.

Absolute values for materials

Displays quantities as absolute values or in percent in the detail application "inventory of permitted materials".

Absolute values for order materials

Displays quantities as absolute values or in percent in the detail applications "selected materials", "composition recipe" and "samples".

Detail application "charging orders“

The detail application "charging orders" shows the list of charging orders.

Charging order

The charging order is indicated with planned order, finished article as well as target quantity and unit.

Planning

Shows the planned dates as well as the melting aggregate

Status

Shows the order status as well as composition status. The composition status results from the last sample taken for this charging order. If this sample is ok ("pass") the composition status is green; if it is not ok ("fail") the status is red.

Price

Shows the average price per kg in € for the material to be produced, the current price per kg in € as well as the total price in €.

Miscellaneous

Displays the scrap rate relating to the selected materials and their properties configured in the material master. The column is highlighted in red if at least one formula for checking the scrap rate is not met. It is highlighted green if all defined formulas are met.

Shows the weight and max. capacity of the melting aggregates used in the order.

Detail application "inventory of permitted materials"

Subject to the entered selection criteria, the list only shows materials that are permitted for the output material or materials that are permitted and have been defined as additional material.

Only will material be taken into account that has been assigned to a casting buffer and to a material type that has been defined as anonymous material (inventory management).

Available quantity

Remaining quantity of a material that is in the batch status "free" and that has not been reserved.

Advised quantity

Remaining quantity of a material that is in the batch status "advised".

Reserved quantity

Remaining quantity of a material that is in the batch status "free" and that has been reserved.

Detail application "selected materials"

This detail application shows the assigned materials.

Input type

Current bottom sump: material that can currently be found in the output buffer of the machine (melting aggregate). The material is determined by the function "adjust bottom sump".

Planned bottom sump material that will be available in the machine's output buffer. The material is determined by the function "adjust bottom sump".

Reserved: material which has been assigned manually using the function "assign material".

BOM item

Planned bottom sump material or the current bottom sump is indicated by the BOM item "0". The material is assigned the BOM item "1" at first. Once charging has been performed and confirmed at the AIP terminal, all materials pertaining to this BOM item are grayed out and the BOM item "1" is blocked. Further materials are now assigned to the next BOM item. Consequently, the current BOM item is blocked every time charging has been performed and the BOM item is increased by one if further materials are added.

Material, designation

Material

Target quantity

Target quantity planned for the production of the finished material. This quantity can be changed by double clicking or the shortcut <Ctrl + Q>. All calculations (formulas, samples: theoretical analysis, charging orders: composition status and scrap rate) are based on the target quantity.

Reserved quantity

Actually reserved quantity that has been reserved using the function "perform reservation" within the batch stock.

Consumed quantity

Actual quantity that has already been consumed and placed in the melting aggregate.

Editor, editing time

The user who has performed the assignment is entered here. The editing time is updated, once the charged material has been confirmed at the AIP terminal.

Detail application "composition recipe"

This detail application shows the composition recipe for the material (finished article) of the selected charging order. The components/elements are indicated with target quantity ("composition recipe" row) as well as with the upper and lower tolerance limits ("upper/lower tolerance limit" rows).

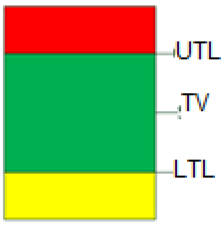
Detail application "formulas“

This detail application shows the composition restrictions for the finished material and the results of the corresponding formula. If the formula is true, the fields are highlighted in green; in any other case they are red.

Formulas are calculated based on the latest sample, i.e. on the current composition of the material (from the analysis process determining the make-up of the sample) and the quantity that is actually entered with this sample at the AIP terminal. All materials assigned after sampling and that have not yet been charged are considered with the relevant target composition and target quantity. All materials that have been charged and confirmed at the AIP terminal after taking the sample are considered with the target composition and the consumed quantity. The calculation is based on all assigned materials, provided that a sample has not yet been taken.

Detail application "Samples"

This detail application shows the samples with the respective sample ID, date and time as well as status (PPKT_NIO=inspection point failed, PPKT_IO=inspection point pass). The share of each element is shown and evaluated (green = value within tolerance limits, yellow = value below the lower tolerance limit, red = value above the upper tolerance limit).



A theoretical analysis is shown additionally referring to the last sample (provided that samples have already been entered) and its current composition as well as to the materials assigned subsequently that might have been charged already - just as it is also the case for the calculation of formulas.

Once the charging order has been released, the theoretical analysis is saved as the "release analysis" ("theoretical analysis for the release").

Toolbar

Start composition

Generate charging order

Provided that no charging order exists for a melting order, this function can be used to generate one.

Start composition for charging order

In case a charging order exists or has been generated, composition is started for the charging order by using the function of the same name. Data of the charging order and of the assigned melting order are shown.

Material reservations

Assign material

Inserts the material or materials selected in "inventory of permitted materials" with the quantity 0 kg for the selected charging order in the "list of charging orders". The material cannot be assigned, provided that the BOM item has already been assigned this material and has not yet been blocked (by confirming the charging process at the AIP terminal).

Change quantity

Changes the target quantity of a selected material. This editing dialog cannot be opened via the quick launch bar. It can only be opened by double clicking a row of the detail application "selected materials" or the shortcut <Ctrl+Q>

The quantity can only be changed if the new quantity is greater than or equal to the quantity that has already been consumed/used. If this is not the case, the user will be informed by an error message.

The quantity cannot be changed if the BOM item is blocked, i.e. the charging process has been confirmed for all materials of this current BOM item at the AIP terminal.

Perform reservation / perform reservation (including material)

This function reserves materials for the selected charging order. Consequently, the required quantities of materials are reserved for this charging order (if available in sufficient quantities) and cannot be used for any other charging order. The reservation applies for all unblocked entries of the order's component list by using the function "perform reservation". As an alternative, it can only be performed for the transferred material by using the function "perform reservation (including material)".

Cancel reservation / cancel reservation (including material)

The reservation of materials is canceled for all unblocked entries of the order's component list using the function "cancel reservation" or only for the selected material using the function "cancel reservation (including material). A reservation can only be canceled completely, provided that material has not yet been consumed.

Adjust bottom sump

This function transfers the material of the batch that is currently within the furnace (current bottom sump) to the currently active composition. In this case, the current bottom sump quantity is also taken over. The theoretical values (planned bottom sump) assumed up to now and resulting from the sequence of orders are overwritten by current information. The bottom sump can no longer be adjusted, once the first charging process has been performed and confirmed at the AIP terminal.

Copy recipe

This function copies the recipe from one charging order to the other. The reservation can be performed afterwards.

Release

Release of charging order

Once the charging order has been released and the materials have been selected, the order is available for execution in production. The "theoretic analysis" presented in the samples is saved as the "release analysis".

Charging list

The charging list can be printed for the released charging order. This list includes the melting order, charging order and a list of required materials.

Further functions

Reuse

If the required material cannot be achieved by adding material (as the melting aggregate's capacity would, for example, be exceeded), the charging order can be assigned to another or new melting order.

Melting report

The melting report outputs all pieces of information referring to the melting process:

- Charging order
- Assigned material and quantity in kg
- Composition recipe
- Current analysis including samples and theoretical analysis